

Selected Topics Of Living Systems HW4

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1 Onsager Relations

1.1 Thermodynamic equations

From the table in the wikipedia page, I chose the Peltier and Seebeck effects. If we look at materials with energy density u , a (to be seen) associated energy current j^u , density n and particle current j^n , Starting from the Gibbs relation

$$ds = \frac{1}{T} du - \frac{\mu}{T} dn \quad (1)$$

where μ is the electrochemical potential, and with the conservation equations

$$\partial_t u + \partial_\alpha j_\alpha^u = 0 \quad \partial_t n + \partial_\alpha j_\alpha^n = 0 \quad (2)$$

differentiating the gibbs relation with respect to time

$$\partial_t s = \frac{1}{T} \partial_t u - \frac{\mu}{T} \partial_t n \quad (3)$$

inserting the conservation equations

$$\dot{S} = \frac{\mu}{T} (\partial_\alpha j_\alpha^n) - \frac{1}{T} (\partial_\alpha j_\alpha^u) = \partial_\alpha \left(\frac{\mu}{T} j_\alpha^n - \frac{1}{T} j_\alpha^u \right) + j_\alpha^u \partial_\alpha \left(\frac{1}{T} \right) - j_\alpha^n \partial_\alpha \left(\frac{\mu}{T} \right) \quad (4)$$

if we define a heat current

$$j^q = j^u - \mu j^n \quad (5)$$

then the equation becomes

$$\dot{S} + \partial_\alpha \left(\frac{j_\alpha^q}{T} \right) = j_\alpha^q \partial_\alpha \left(\frac{1}{T} \right) - \frac{1}{T} j_\alpha^n \partial_\alpha \mu \quad (6)$$

we can identify force current pairs like we did in the lecture. The heat current is associated with a force which depends on the temperature gradient, and the particle current is associated with a force which depends on both on temperature gradient and the electrochemical potential gradient. If we assume the dynamics are linear we get

$$\begin{pmatrix} j_\alpha^n \\ j_\alpha^q \end{pmatrix} = \begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{pmatrix} \begin{pmatrix} \partial_\alpha (\mu/T) \\ \partial_\alpha (1/T) \end{pmatrix} \quad (7)$$

Since both the currents are odd under time reversal and the forces are even, we know the coupling is dissipative, therefore $\gamma_{12} = \gamma_{21}$. Looking at the cross terms

$$j_\alpha^n = \dots + \gamma_{12} \partial_\alpha (1/T) \quad (8)$$

$$j_\alpha^q = \dots + \gamma_{12} \partial_\alpha (\mu/T) \quad (9)$$

The first equation tells us that a gradient in temperature will create a current of charge carriers. If the system isn't connected in a loop, this will generate an electromotive force which we can then use to write

$$\varepsilon = S \nabla T \quad (10)$$

which is the more commonly known form of the Seebeck effect. The second equation tells us there's a connection between the concentration of charge carriers and the heat of the system. A more common form

for the Peltier effect is

$$j_\alpha^q = \Pi j_\alpha - \kappa \nabla T \quad (11)$$

where Π are Peltier coefficients and $j_\alpha = q j_\alpha^n$ is the electric current. This equation states the same as we did in the 2nd equation, just in a clearer form.

1.2 Entropy Production Rate

Since the system has two components and is isotropic and passive, we have the conservation equation

$$\partial_t \rho_i + \partial_\alpha (\rho_i v_\alpha^i) = 0 \quad ; \quad i = 1, 2 \quad (12)$$

To move to the center of mass frame we'll define the total density as $\rho = \rho_1 + \rho_2$ and the center of mass velocity

$$v_\alpha = \frac{\rho_1 v_\alpha^1 + \rho_2 v_\alpha^2}{\rho} \quad (13)$$

then

$$\rho_1 v_\alpha^1 = \rho_1 v_\alpha + j_\alpha \quad \rho_2 v_\alpha^2 = \rho_2 v_\alpha - j_\alpha \quad (14)$$

We'll define the chemical potential as expected

$$\mu_i \equiv \frac{\delta F}{\delta \rho_i} \quad (15)$$

where F is the free energy. We'll assume the system is coupled to a heat reservoir as we did in the lecture, therefore

$$\dot{S} = -\frac{1}{T} \frac{dF}{dt} \quad (16)$$

calculating the derivative as we did in the lecture

$$\frac{dF}{dt} = \int d\mathbf{r} (\mu_1 \partial_t \rho_1 + \mu_2 \partial_t \rho_2) \quad (17)$$

inserting the continuity equation

$$\frac{dF}{dt} = - \int d\mathbf{r} (\mu_1 \partial_\alpha (\rho_1 v_\alpha^1) + \mu_2 \partial_\alpha (\rho_2 v_\alpha^2)) \quad (18)$$

performing integration by parts

$$\frac{dF}{dt} = \int d\mathbf{r} (\rho_1 v_\alpha^1 \partial_\alpha \mu_1 + \rho_2 v_\alpha^2 \partial_\alpha \mu_2) \quad (19)$$

inserting the center of mass definitions

$$\frac{dF}{dt} = \int d\mathbf{r} ((\rho_1 v_\alpha + j_\alpha) \partial_\alpha \mu_1 + (\rho_2 v_\alpha - j_\alpha) \partial_\alpha \mu_2) \quad (20)$$

rearranging

$$\frac{dF}{dt} = \int d\mathbf{r} [v_\alpha (\rho_1 \partial_\alpha \mu_1 + \rho_2 \partial_\alpha \mu_2) + j_\alpha \partial_\alpha (\mu_1 - \mu_2)] \quad (21)$$

noticing the first term is very similar to the form we saw in the lecture we can write

$$T\dot{S} = \int d\mathbf{r} [\sigma_{\alpha\beta} v_{\alpha\beta} + j_{\alpha} (-\partial_{\alpha} (\mu_1 - \mu_2))] \quad (22)$$

The 2nd element is therefore the one we're interested in. The new current j_{α} is the relative movement of the materials with respect to their shared center of mass, and the new force is the gradient of the difference in chemical potential between the materials. Doing a similar process to the lecture we find

$$j_{\alpha} = -\gamma \partial_{\alpha} (\mu_1 - \mu_2) \quad (23)$$

plugging it back into the entropy Production

$$T\dot{S} = \int d\mathbf{r} [\sigma_{\alpha\beta} v_{\alpha\beta} + \gamma |\partial_{\alpha} (\mu_1 - \mu_2)|^2] \quad (24)$$

Qualitatively, the second element being the square of the difference between chemical potentials means the two material will diffuse into each other to lower the free energy, with the diffusion happening into one material or the other not being preferred.

2 Spontaneous Flow Transition

Recalling the equations for the force and rate of change of the directional vector:

$$\partial_\beta \sigma_{\alpha\beta}^{tot} = 0 \quad (\partial_t + v_\beta \partial_\beta) p_\alpha = \frac{h_\parallel}{\gamma_1} p_\alpha + \frac{K}{\gamma_1} \partial_{\beta\beta} p_\alpha - (\omega_{\alpha\beta} + \nu v_{\alpha\beta}) p_\beta \quad (25)$$

In our case $\nu = 0$ which cancels out the 2nd element inside the parentheses in the last element. The entire parentheses describes the rotation and alignment of the directional vector with the flow. Removing the element depending on ν removes the alignment element so the directional vector now only spins as a result of the flow and doesn't seek to align with it. To begin the linear stability test we'll write the equations for a small perturbation in the angle of p for which $\mathbf{p} = (1, \theta)$. Inserting this into the equation for the vector gives us

$$(\partial_t + v_\beta \partial_\beta) p_\alpha = \frac{h_\parallel}{\gamma_1} p_\alpha + \frac{K}{\gamma_1} \partial_{\beta\beta} p_\alpha - \omega_{\alpha\beta} p_\beta \quad (26)$$

dropping elements in 2nd order for perturbation and the lagrange multiplier (since for small perturbation it's effectively unneeded) Inserting this into the equation for the vector gives us

$$\partial_t \theta = \frac{K}{\gamma_1} \partial_{\beta\beta} \theta - \omega_{yx} \quad (27)$$

where on the left hand side the time derivative of 1 is 0, and on the right hand side the angular velocity is an order of perturbation so it couples only with the constant in the vector in first order which also imposes the selection of indices. We can express the vorticity element as

$$\omega_{yx} = \frac{\partial_y v_x - \partial_x v_y}{2} = \frac{1}{2} \partial_{\alpha\alpha} \psi \quad (28)$$

which gives us

$$\partial_t \theta = \frac{K}{\gamma_1} \partial_{\beta\beta} \theta - \frac{1}{2} \partial_{\alpha\alpha} \psi \quad (29)$$

performing a fourier transform gives us

$$s\hat{\theta} = -\frac{K}{\gamma_1} q^2 \hat{\theta} + \frac{1}{2} q^2 \hat{\psi} \quad (30)$$

Next we need to find a way to connect ψ to θ . Using the force equation

$$\partial_\beta \sigma_{\alpha\beta}^{tot} = 0 \quad (31)$$

we can take the rotor of this and using the definition from the lecture we get

$$\partial_x \partial_\beta \sigma_{\alpha\beta}^{tot} - \partial_y \partial_\beta \sigma_{\alpha\beta}^{tot} = \eta \nabla^2 \omega_z \quad (32)$$

for the viscous term, and if we fourier transform it it becomes

$$-\eta q^2 \omega_z \quad (33)$$

As for the active term $\zeta \Delta \mu Q_{\alpha\beta}$ since we're in linear order only $Q_{xy} = Q_{yx} \approx \theta$ matters so the active force is

$$f_\alpha = -\zeta \Delta \mu \partial_\beta \theta \quad (34)$$

whose rotor is

$$\partial_x f_y - \partial_y f_x = -\zeta \Delta \mu (\partial_x^2 - \partial_y^2) \theta \quad (35)$$

and after a fourier transform it becomes

$$\zeta \Delta \mu (q_x^2 - q_y^2) \hat{\theta} \quad (36)$$

we can rewrite the qs using

$$q_x^2 - q_y^2 = q^2 (\cos^2 \phi - \sin^2 \phi) = q^2 \cos(2\phi) \quad (37)$$

so the fourier transformed element becomes

$$\zeta \Delta \mu q^2 \cos(2\phi) \hat{\theta} \quad (38)$$

lastly the rotation and alignment terms we have

$$\frac{1}{2} (p_\alpha h_\beta - h_\alpha p_\beta) \quad (39)$$

recalling a definition from lecture 6

$$h_\alpha = h_{||} p_\alpha + K \partial_{\beta\beta} p_\alpha \quad (40)$$

and using the small perturbation we find

$$h_y \approx K \partial_{\beta\beta} \theta \quad (41)$$

and h_x is eliminated by the lagrange multiplier elements. Doing the same rotor trick we get an element proportional to the 4th derivative of θ which in fourier space is

$$-\frac{K}{2} q^4 \hat{\theta} \quad (42)$$

putting the pieces together we find

$$-\eta q^2 \omega_z + \zeta \Delta \mu q^2 \cos(2\phi) \hat{\theta} - \frac{K}{2} q^4 \hat{\theta} = 0 \quad (43)$$

inserting $\omega_z = q^2 \hat{\psi}$

$$-\eta q^4 \hat{\psi} + \zeta \Delta \mu q^2 \cos(2\phi) \hat{\theta} - \frac{K}{2} q^4 \hat{\theta} = 0 \quad (44)$$

dividing by q^2 (assuming the wavevector isn't zero)

$$-\eta q^2 \hat{\psi} + \zeta \Delta \mu \cos(2\phi) \hat{\theta} - \frac{K}{2} q^2 \hat{\theta} = 0 \quad (45)$$

rearranging

$$\hat{\psi} = \frac{\zeta \Delta \mu}{\eta q^2} \cos(2\phi) \hat{\theta} - \frac{K}{2\eta} \hat{\theta} \quad (46)$$

inserting this into equation 30

$$s\hat{\theta} = -\frac{K}{\gamma_1} q^2 \hat{\theta} + \frac{1}{2} q^2 \left(\frac{\zeta \Delta \mu}{\eta q^2} \cos(2\phi) \hat{\theta} - \frac{K}{2\eta} \hat{\theta} \right) \quad (47)$$

simplifying to

$$s\hat{\theta} = -\frac{K}{\gamma_1}q^2\hat{\theta} + \frac{\zeta\Delta\mu}{2\eta}\cos(2\phi)\hat{\theta} - \frac{Kq^2}{4\eta}\hat{\theta} \quad (48)$$

which gives us the dispersion relation

$$s = \frac{\zeta\Delta\mu}{2\eta}\cos(2\phi) - Kq^2\left(\frac{1}{\gamma_1} + \frac{1}{4\eta}\right) \quad (49)$$

which to be unstable must be positive for a small q , so we'll take the limit $q \rightarrow 0$ which gives us the requirement

$$\frac{\zeta\Delta\mu}{2\eta}\cos(2\phi) > 0 \quad (50)$$

so the critical wavevector is

$$q_c^2 = \frac{\frac{\zeta\Delta\mu}{2\eta}\cos(2\phi)}{K\left(\frac{1}{\gamma_1} + \frac{1}{4\eta}\right)} \quad (51)$$